

FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10: SUB-STRAND 2.7 VOLUME AND CAPACITY — FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

This is your Final Explanation for the unit. Use what you have learned in all eight lessons about the Kericho tea farmers' three containers: the cylindrical drum (A), the conical sisal cover (B) and the metal frustum bucket (C). Write your answers in the space under each prompt. Use full sentences, show every step of any calculation, include units, and refer to the measurements from the lessons where a question asks you to. Use the formulas and ideas we developed in class, and explain why each one works as well as what it is.

Part 1: The three containers and the 3 : 1 pattern

In Lesson 1 the three Kericho containers were filled to the brim. The drum (A) held about 9.4 L, the conical cover (B) about 3.1 L and the metal bucket (C) about 6.5 L, even though all three have roughly the same height and opening diameter. (a) Work out the ratio of the capacity of A to the capacity of B and say what it suggests. (b) Describe the pouring experiment from Lesson 2 that showed this ratio, and say what you saw. (c) Starting from $V = \pi r^2 h$ for the cylinder, show how you get the volume formula for the cone. (d) Explain in your own words why the cone holds less than the cylinder even though they have the same height and opening width.

(a) $9.4 \div 3.1 \approx 3.03$, so the ratio is about 3 : 1. The cylindrical drum holds about three times as much as the conical cover. (b) We filled a transparent cone with water and poured it into a matching cylinder with the same base radius r and the same height h . The cylinder was exactly full after the third pour, so $V(\text{cylinder}) = 3 \times V(\text{cone})$. (c) Since $V(\text{cylinder}) = 3 \times V(\text{cone})$, $V(\text{cone}) = \frac{1}{3} \times V(\text{cylinder}) = \frac{1}{3}\pi r^2 h$. (d) The cylinder keeps the same full circle of area πr^2 at every level from base to top. The cone narrows steadily to a point, so most of its cross-sections are much smaller circles. It therefore fills only one-third of the space the cylinder fills, even though it has the same height and opening width.

Part 2: Where does the metal bucket fit?

Container C, the metal bucket, is a frustum of a cone. (a) Where does its capacity (about 6.5 L) sit compared with the cone (about 3.1 L) and the cylinder (about 9.4 L)? (b) Explain how a frustum is related to two cones. (c) Write the volume formula for a frustum of a cone, where R is the top radius, r is the bottom radius and h is the vertical height, and state which two cone volumes are subtracted to obtain it. (d) Explain why the bucket holds more water than the cone even though both have the same height and top opening.

(a) The bucket's 6.5 L is more than the cone's 3.1 L but less than the cylinder's 9.4 L, so it sits between them. (b) A frustum is a large cone with a smaller cone cut off at the narrow end. The remaining piece is the frustum. (c) $V(\text{frustum}) = \frac{1}{3}\pi h(R^2 + Rr + r^2)$. It comes from $V(\text{large cone}) - V(\text{small cone})$. We used similar triangles to eliminate the unknown full height of the large cone, and expanding the subtraction gives the three terms R^2 , Rr and r^2 . (d) Cutting off the narrow tip of the cone removes very little volume, because the tip is so thin. The frustum keeps a wide top (the opening) and only narrows to a smaller circle at the bottom, so it keeps much more of the middle volume than a cone that narrows to a point. It therefore holds more than the cone, but because it still narrows from top to bottom it cannot hold as much as the cylinder, which stays full-width all the way.

Part 3: Pyramids, frustums of pyramids and units

(a) In Lesson 3 you filled a square-based pyramid with rice and poured it into a cuboid with the same base and height. State what you found and write the volume formula for the pyramid. (b) Explain what the cone-cylinder result and the pyramid-cuboid result have in common. (c) Write the volume formula for a frustum of a square pyramid in terms of the areas A_1 and A_2 of its two parallel faces and the height h , and explain what is the same and what is different between this and the frustum of a cone. (d) A container has a volume of $9\,400\text{ cm}^3$. Convert this to litres, and give the conversion relationships you used.

(a) It took 3 full pyramids to fill the cuboid, so $V(\text{pyramid}) = \frac{1}{3} \times \text{base area} \times \text{height}$. (b) In both cases the pointed solid (the cone or the pyramid) holds exactly one-third of the solid with the same base and height that surrounds it (the cylinder or the cuboid). This is true whether the base is circular or square. (c) $V = \frac{1}{3}h(A_1 + A_2 + \sqrt{A_1A_2})$. The structure is the same as for the frustum of a cone: one-third times the height times the sum of the two end areas and their geometric mean. What differs is how the areas are calculated: for a cone frustum $A = \pi r^2$, and for a square pyramid frustum $A = s^2$. (d) $1\text{ cm}^3 = 1\text{ mL}$ and $1\,000\text{ cm}^3 = 1\text{ L}$ (also $1\text{ m}^3 = 1\,000\text{ L}$). So $9\,400\text{ cm}^3 = 9\,400\text{ mL} = 9.4\text{ L}$.

Part 4: Designing a container of exact capacity

A Kericho farmer wants a new frustum-shaped metal bucket that holds 10 litres. Available sheet metal fixes the vertical height at $h = 30$ cm and the bottom radius at $r = 8$ cm. (a) Convert the target capacity to cm^3 . (b) Substitute into the frustum formula and write an equation for the top radius R . (c) Solve for R , giving your answer to 1 decimal place, and say whether it is a sensible size for a bucket with $r = 8$ cm. (d) If you built a model bucket like this and found that it held slightly more or less than 10 L when filled with water, give two possible reasons for the difference, based on your project in Lesson 7.

(a) $10 \text{ L} = 10 \times 1\,000 \text{ cm}^3 = 10\,000 \text{ cm}^3$. (b) $10\,000 = \frac{1}{3} \times \pi \times 30 \times (R^2 + 8R + 64)$, which simplifies to $10\,000 = 10\pi(R^2 + 8R + 64)$. Then $R^2 + 8R + 64 = 1\,000 \div \pi \approx 318.3$, so $R^2 + 8R - 254.3 = 0$. (c) Using the quadratic formula, $R = (-8 + \sqrt{64 + 1017.2}) \div 2 = (-8 + 32.88) \div 2 \approx 12.4$ cm. Check: $12.4^2 + 8 \times 12.4 + 64 = 153.76 + 99.2 + 64 = 316.96$, and $10\pi \times 316.96 \approx 9\,957 \text{ cm}^3 \approx 10 \text{ L}$. A top radius of 12.4 cm (diameter about 25 cm) is wider than the bottom radius of 8 cm, so the bucket opens upwards like the Kericho bucket, and this is a sensible size. (d) Possible reasons: small errors in measuring the height or radii (even 1 cm makes a calculable change in capacity); the cut and folded cardboard or metal not forming a perfect shape; water not filled exactly to the brim or some left behind when poured. These explain the percentage discrepancy between theoretical and experimental capacity.

Part 5: Answering the driving question

Our driving question was: Why do the tea farmers' containers, a cone, a cylinder and a bucket, look nearly the same size but hold such different amounts of water, and how can we use mathematics to design a frustum-shaped container that holds exactly the capacity we need? Write a clear explanation of 6 to 8 sentences. Include (i) the relationship between the cylindrical drum and the conical cover, (ii) why the frustum bucket falls between them, (iii) how the formulas let a farmer design a container, and (iv) your answer to the KICD question: why is the knowledge of volume and capacity useful?

The drum, the conical cover and the bucket look the same size because they have roughly the same height and opening diameter, but their insides are shaped differently, so the area of each cross-section builds up differently from base to top. The cylinder keeps a full circle of area πr^2 at every level, so it holds the most. The cone narrows steadily to a point, so it holds exactly one-third of the matching cylinder, $V = \frac{1}{3}\pi r^2 h$, which matches 3.1 L compared with 9.4 L. The frustum bucket is a large cone with a small cone removed, $V = \frac{1}{3}\pi h(R^2 + Rr + r^2)$. It keeps a wide top opening and narrows only to a smaller circle at the bottom, not to a point, so it holds more than the cone but less than the cylinder (about 6.5 L). The same idea works for pyramids and their frustums, using $V = \frac{1}{3}h(A_1 + A_2 + \sqrt{A_1 A_2})$. A farmer who wants a container of exactly 10 L can fix the height and one radius, then solve the formula for the other radius. Knowing volume and capacity is useful because it lets us compare, predict and design containers such as tanks, cooking pots, grain bins and tea buckets, saving materials and money and avoiding guesswork.

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Use of the anchoring phenomenon and data	Uses the Kericho containers and the measured capacities (9.4 L, 3.1 L, 6.5 L) correctly and consistently to support every explanation.	Refers to the containers and most of the data correctly, with minor gaps in linking data to explanations.	Mentions the containers but uses data incorrectly or not at all.
Correct formulas and relationships	States and applies the cone, pyramid, cone-frustum and pyramid-frustum formulas correctly, and explains where the one-third factor and the frustum formula come from.	States the formulas correctly but gives only partial explanation of where they come from.	Formulas are missing, incorrect or only memorised with no reasoning.
Calculation and unit conversion	Calculations are accurate and clearly laid out, with correct conversions between cm^3 , mL and L and correct solving for R.	Method is correct with small arithmetic or unit slips.	Method is unclear or contains major errors in calculation or units.
Scientific explanation of why shapes hold different amounts	Explains clearly, using cross-sections and the cone-removed-from-cone idea, why the cylinder holds most, the cone least and the frustum in between.	Gives a mostly correct explanation, but relies on general statements.	Explanation is incomplete or only describes what the containers look like.
Application, design and communication	Designs a container for a target capacity, identifies realistic sources of error and explains the usefulness of volume and capacity in clear, well-organised sentences.	Completes the design task and gives some sources of error and uses, with minor lapses in clarity.	Design is incomplete and the uses of volume and capacity are unclear.

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Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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